

Keplerian Orbital Equations

Radius

$$r = \frac{p}{e \cos(\nu) + 1}$$

Radius of Apoapsis

$$r_a = a(e + 1)$$

Characteristic Energy

$$C_3 = -\frac{\mu}{a}$$

Specific Angular Momentum

$$h = \sqrt{\mu} \sqrt{p} = r v \cos(\gamma)$$

True Anomaly Rate

$$\dot{\nu} = \frac{a^2 n \sqrt{1 - e^2}}{r^2}$$

True Anomaly

$$\nu = \operatorname{atan}_2(\sqrt{1 - e^2} \sin(E), -e + \cos(E))$$

Flight Path Angle - Parabolic

$$\gamma = \frac{\nu}{2}$$

Perifocal Radius

$$\mathbf{r}_{PQW} = \begin{bmatrix} \frac{p \cos(\nu)}{e \cos(\nu) + 1} \\ \frac{p \sin(\nu)}{e \cos(\nu) + 1} \\ 0 \end{bmatrix}$$

Radius Vector: Inertial

$$r_{IJK} = [\frac{IJK}{PQW}] \mathbf{r}_{PQW}$$

Equinoctial Eccentricity Sine Term

$$g = e \sin(\Omega + \omega)$$

Delaunay L

$$L_d = \sqrt{\mu} \sqrt{a}$$

Semi-Latus Rectum

$$p = a(1 - e^2)$$

Velocity (elliptical)

$$v = \sqrt{\frac{2\mu}{r} - \frac{\mu}{a}}$$

Velocity (circular)

$$v_c = \frac{\sqrt{\mu}}{\sqrt{r}}$$

Mean Anomaly (Elliptical)

$$M = E - e \sin(E)$$

Radial Rate

$$\dot{r} = \frac{\dot{\nu} e r \sin(\nu)}{e \cos(\nu) + 1}$$

Flight Path Angle

$$\gamma = \operatorname{atan}_2(e \sin(E), 1 - e^2)$$

Hyperbolic Anomaly

$$H = \operatorname{atan}_2(\sqrt{e^2 - 1} \sinh(\nu), e + \cosh(\nu))$$

Perifocal Velocity

$$\mathbf{v}_{PQW} = \begin{bmatrix} -\frac{\sqrt{\mu} \sin(\nu)}{\sqrt{p}} \\ \frac{\sqrt{\mu}(e + \cos(\nu))}{\sqrt{p}} \\ 0 \end{bmatrix}$$

Velocity Vector: Inertial

$$v_{IJK} = [\frac{IJK}{PQW}] \mathbf{v}_{PQW}$$

Equinoctial Inclination Cosine Term

$$h = \cos(\Omega) \tan\left(\frac{i}{2}\right)$$

Delaunay H

$$H_d = \sqrt{\mu} \sqrt{a(1 - e^2)} \cos(i)$$

Semi-Minor Axis

$$b = a \sqrt{1 - e^2}$$

Vis-Viva

$$v = \sqrt{\mu} \sqrt{\frac{2}{r} - \frac{1}{a}}$$

Escape Velocity

$$v_{esc} = \frac{\sqrt{2} \sqrt{\mu}}{\sqrt{r}}$$

Mean Motion

$$n = \sqrt{\mu} \sqrt{\frac{1}{a^3}}$$

Argument of Latitude

$$u = \nu + \omega$$

Semi-Latus Rectum (Parabolic)

$$p = \frac{h^2}{\mu}$$

Mean Anomaly (Hyperbolic)

$$M = -H + e \sinh(H)$$

Perifocal to Inertial Rotation Matrix

$$[\frac{IJK}{PQW}] = \begin{bmatrix} -\sin(\Omega) \sin(\omega) \cos(i) + \cos(\Omega) \cos(\omega) & -\sin(\Omega) \cos(\omega) \cos(i) - \sin(\omega) \cos(\Omega) & \sin(\Omega) \sin(i) \\ \sin(\Omega) \cos(\omega) + \sin(\omega) \cos(\Omega) \cos(i) & -\sin(\Omega) \sin(\omega) + \cos(\Omega) \cos(\omega) \cos(i) & -\sin(i) \cos(\Omega) \\ \sin(\omega) \sin(i) & \sin(i) \cos(\omega) & \cos(i) \end{bmatrix}$$

Two Body Differential Equation

$$\ddot{\mathbf{r}} = -\frac{\mu}{r^3} \mathbf{r}_{IJK}$$

Equinoctial Inclination Sine Term

$$k = \sin(\Omega) \tan\left(\frac{i}{2}\right)$$

Radius of Periapsis

$$r_p = a(1 - e)$$

Specific Orbital Energy

$$\varepsilon = -\frac{\mu}{2a} = -\frac{\mu}{r} + \frac{v^2}{2}$$

Hyperbolic Excess Velocity

$$v_\infty = \sqrt{\mu} \sqrt{-\frac{1}{a}}$$

Period

$$T = \frac{2\pi \sqrt{a^3}}{\sqrt{\mu}}$$

Eccentric Anomaly

$$E = \operatorname{atan}_2(\sqrt{1 - e^2} \sin(\nu), e + \cos(\nu))$$

Parabolic Anomaly

$$B = \operatorname{atan}(\sin(\frac{\nu}{2}))$$

Hyperbolic Turning Angle

$$\epsilon = 2 \operatorname{asin}\left(\frac{1}{e}\right)$$

Equinoctial Eccentricity Cosine Term

$$f = e \cos(\Omega + \omega)$$

Mean Longitude

$$L = \Omega + \nu + \omega$$

Sources

Bates, Mueller, White — *Fundamentals of Astrodynamics*

- **Escape Velocity** — p. 35: Eq 1.9-2
- **Period** — p 33: Eq 1.7-9
- **Radius** — p. 20: Eq 1.5-4
- **Semi-Latus Rectum** — p. 24: Eq 1.5-6
- **Semi-Minor Axis** — p. 15: Eq 1.4
- **Velocity (circular)** — p. 34: Eq 1.8-2

Jacob Williams — *Modified Equinoctial Elements*

- **Equinoctial Eccentricity Cosine Term** — <https://degenerateconic.com/modified-equinoctial-elements.html>
- **Equinoctial Eccentricity Sine Term** — <https://degenerateconic.com/modified-equinoctial-elements.html>
- **Equinoctial Inclination Cosine Term** — <https://degenerateconic.com/modified-equinoctial-elements.html>
- **Equinoctial Inclination Sine Term** — <https://degenerateconic.com/modified-equinoctial-elements.html>
- **Mean Longitude** — <https://degenerateconic.com/modified-equinoctial-elements.html>

Vallado — *Fundamentals of Astrodynamics and Applications* — 4th Edition

- **Argument of Latitude** — 2-90 p. 102
- **Characteristic Energy** — p. 27
- **Delaunay L** — 2-102
- **Delaunay H** — 2-102
- **Eccentric Anomaly** — Simplified from other expressions
- **Flight Path Angle** — Simplified from other expressions

- **Flight Path Angle - Parabolic** — p. 2
- **Hyperbolic Anomaly** — Simplified from other expressions
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- **Mean Motion** — p. 45: Eq 2-5
- **Parabolic Anomaly** — p. 2
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- **True Anomaly Rate** — p. 2
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- **Velocity Vector: Inertial** — Algorithm 10, page 119
- **Vis-Viva** — p. 27, Eq. 1-22

Note on atan2: This sheet uses $\text{atan2}(y, x)$ (sine term/ y first, cosine term/ x second) to preserve quadrant.